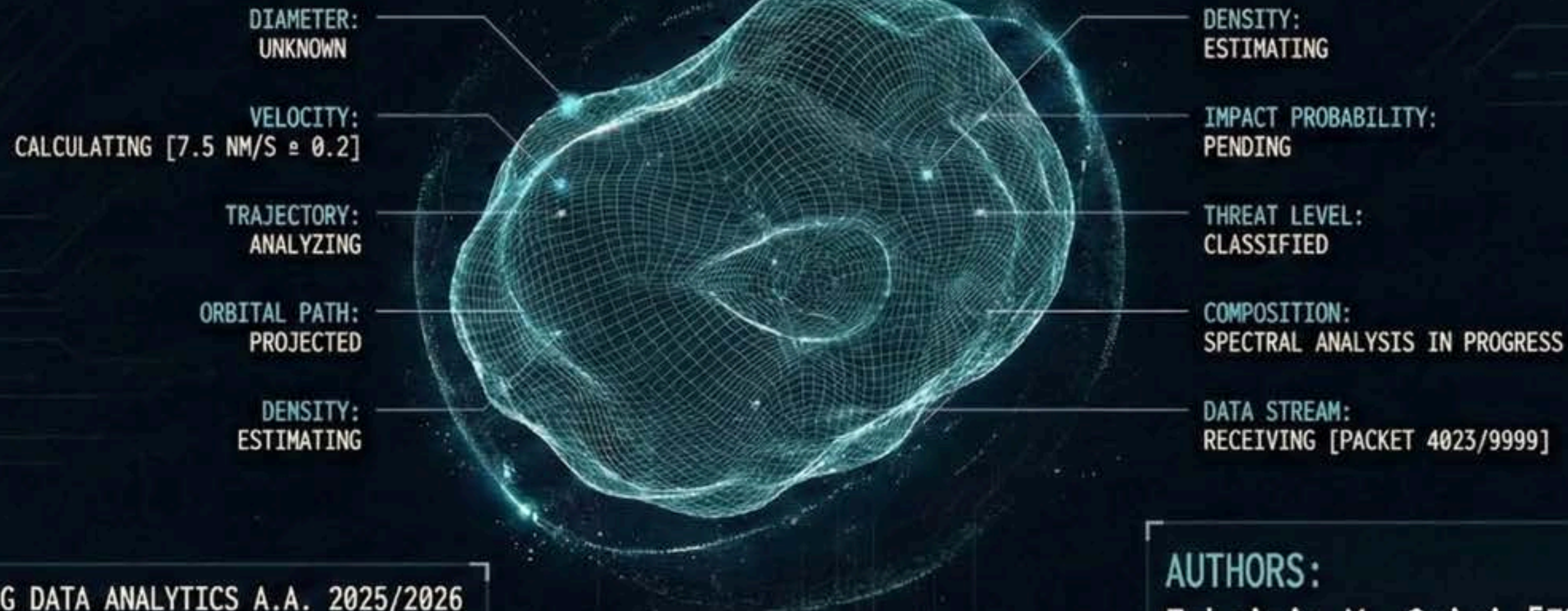


NASA ASTEROIDS CLASSIFICATION AI FOR PLANETARY DEFENSE



PROJECT: BIG DATA ANALYTICS A.A. 2025/2026
SCOPE: SPACE / DEFENSE
INSTITUTION:
Politecnico di Bari
Dipartimento di Meccanica, Matematica e Management

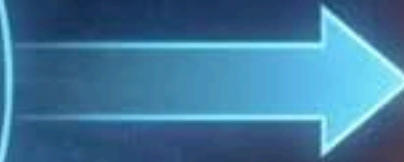
AUTHORS:

Fabrizio M. Crimi [ID: 593588]
Michele Falconetti [ID: 596924]
Michele Storelli [ID: 598652]

THREAT CONTEXT: NEAR-EARTH OBJECTS (NEO)



Energy: 10–15 Megatons TNT.
Devastation: 2,000 km².



Energy: 500 Kilotons TNT.
Result: 1,500+ Injured.



DEFINITION:

NEO: Perihelion < 1.3 AU

● **PHO:** Potentially Hazardous Object

CURRENT STATUS:



Monitored Objects: 30,000+ (NASA CNEOS/PDCO)



The Challenge: 95% Safe / 5% Hazardous



Mission: Anomaly detection in imbalanced datasets

THE CHALLENGE OF EARLY DETECTION

INPUT: TELESCOPE ARRAY



Data Volume: Terabytes of daily celestial scans.

PROCESSING: NEURAL NETWORK



Objective: Filter priority observation targets.
Mechanism: Automated classification systems.

CASE STUDIES:

- 1. 99942 Apophis:** High risk reclassified via better data.
- 2. DART Mission (2022):** Kinetic deflection proven. Success relies on early identification.

INTELLIGENCE SOURCE: DATASET

SOURCE METADATA

Origin: NASA Official Databases via Kaggle

Period: 1910–2024

Volume: 90,000+ Records

Format: CSV (Structured)

NASA - Nearest Earth Objects

A cumulative data for Nearest Earth Objects by NASA



id	name	# est_diameter...	# est_diameter...	# relative_velo...	# miss_distance	orbiting_body	sentry_object	# absolute_ma...	hazardous
2162635	162235 (2000 SST64)	1.1982700007	2.6794149958	13569.2492241012	54839744.08294605	Earth	False	16.73	False
2277475	277475 (2005 WK4)	0.26S8	0.5594486694	73583.7286894981	61438126.32333093	Earth	False	20.0	True
2512244	S12244 (2015 YE18)	0.7220295677	1.6145071127	114258.2521990512	49798724.94445679	Earth	False	17.83	False
3396000	(2012 BV13)	0.096555147	0.2577948049	24764.3031380016	25444577.72075825	Earth	False	22.2	False
3667127	(2014 GEYS)	0.2530088879	0.3702167609	42797.3337887224	46575557.00130072	Earth	False	20.09	True
54188966	(2021 GYY3)	0.0665543223	0.0812093344	34297.5877883029	40555661.22792288	Earth	False	24.32	False
54189957	(2021 PY40) AF2)	0.1116148941	3.8753743344	27592.5787789963	20065921.41664897	Earth	False	20.95	False
54230078	(2021 XD6) AF2)	0.0033278866	0.0119135161	57544.4700827552	55115019.25807114	Earth	False	23.49	False

TARGET ACQUISITION: BINARY CLASSIFICATION

VARIABLE: is_potentially_hazardous_asteroid

LOGIC: PHYSICAL THREAT

TRUE (1): Large diameter + Earth-intersecting orbit

FALSE (0): No immediate threat

NOTE: The primary target of this AI model



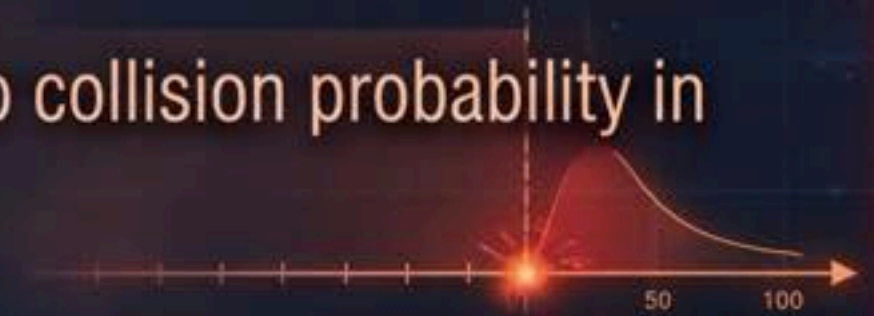
VARIABLE: sentry_object

LOGIC: PROBABILITY OF IMPACT

TRUE (1): Non-zero collision probability in next 100 years

FALSE (0): Zero collision probability

DISTINCTION: A 'Sentry' object is an extremely rare subset of 'Hazardous'. An object can be Hazardous (theoretically) without being a Sentry (imminent)



INPUT PARAMETERS: KEY FEATURES

ABSOLUTE MAGNITUDE (H)



Intrinsic luminosity at 1 AU.
Proxy for size.

ESTIMATED DIAMETER



Min/Max confidence interval
derived from luminosity.

RELATIVE VELOCITY



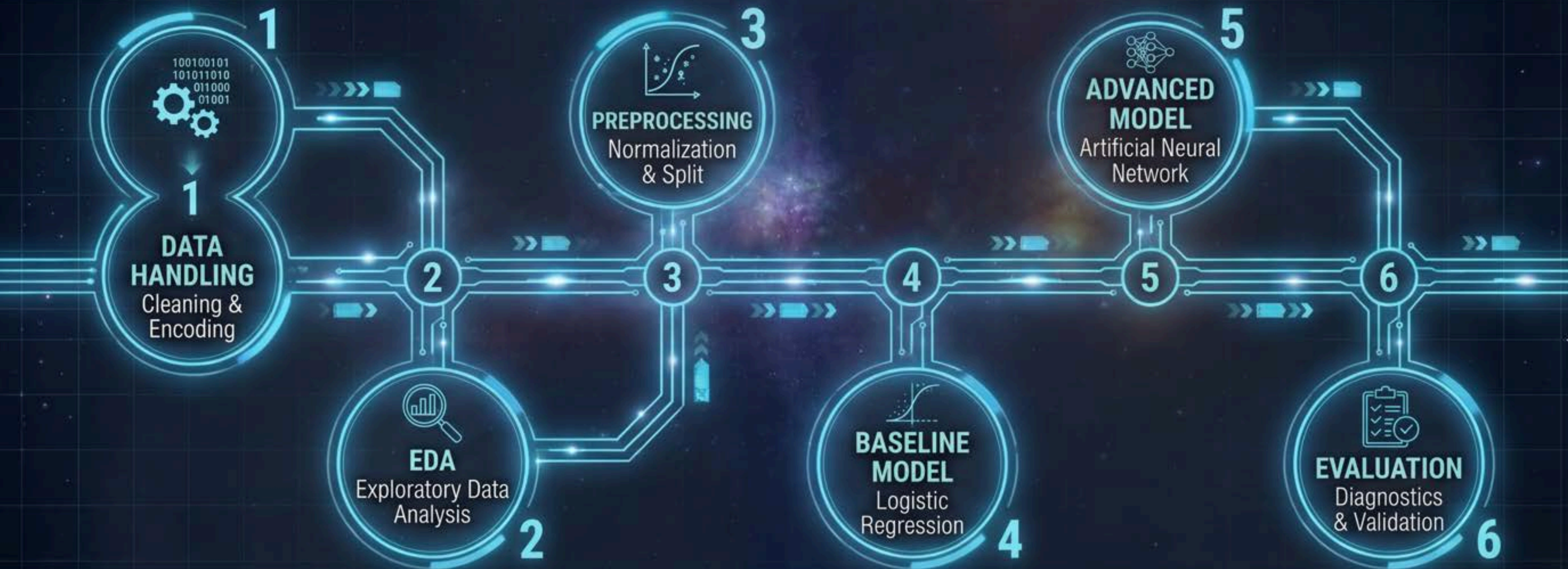
Instantaneous speed at
close approach. Critical for
Kinetic Energy ($E = 1/2 mv^2$).

MISS DISTANCE (MOID)



Minimum Orbit Intersection
Distance. Direct proxy for
collision probability.

THE ANALYTICS PIPELINE

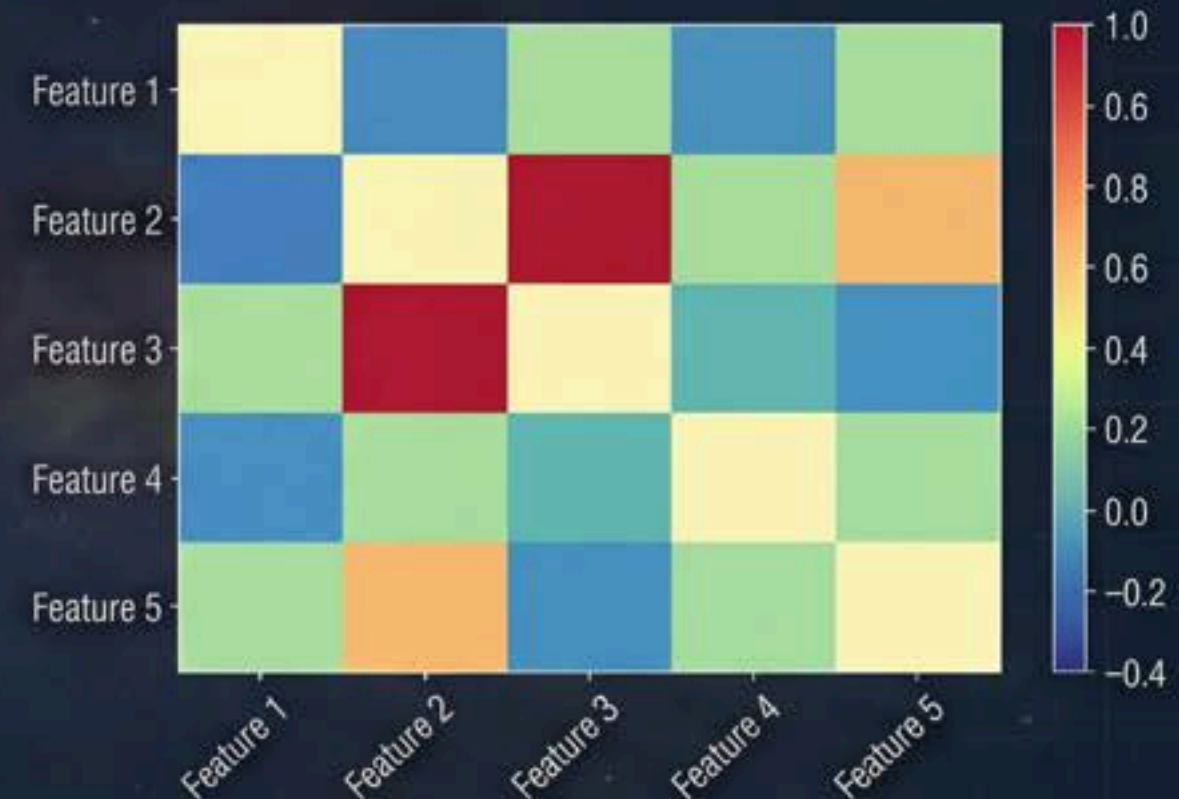


PHASE 1 & 2: HANDLING & RECONNAISSANCE

PHASE 1: CLEANING & ENCODING

- ✓ Drop Non-Predictive Metadata: [ID], [Name].
- ✓ Impute Missing Values: Statistical replacement of NaN.
- ✓ Encoding: Boolean (True/False) -> Binary (1/0).

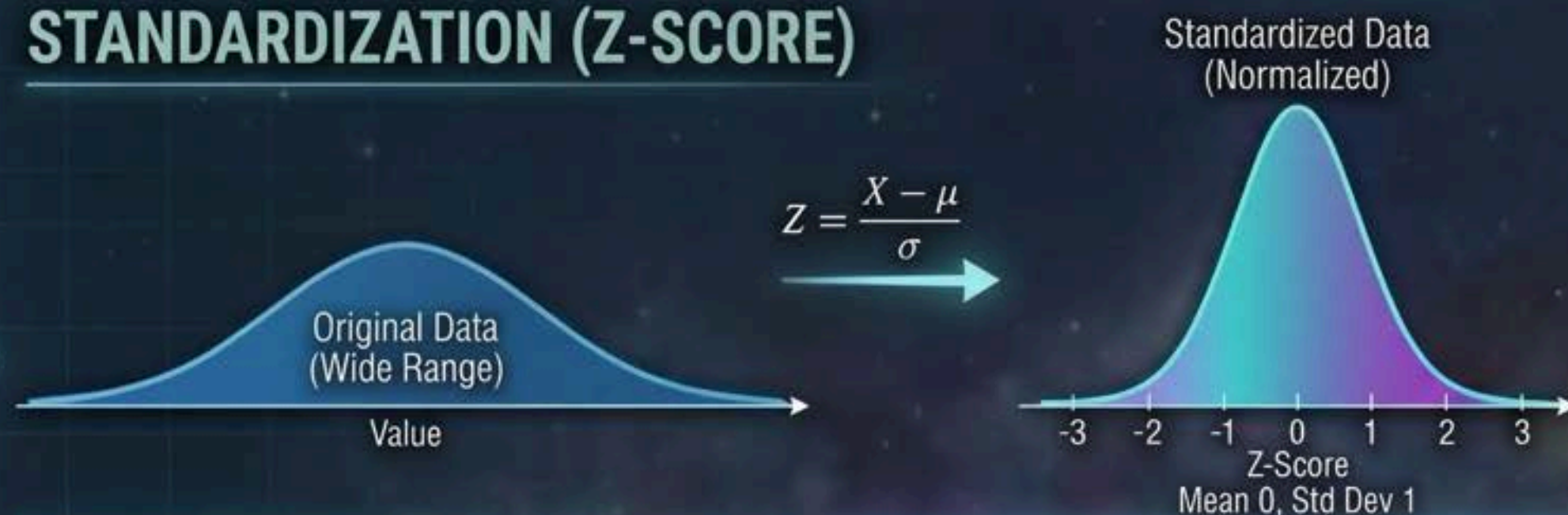
PHASE 2: ANALYSIS (EDA)



- ☰ **Zero Variance Check:** Removal of static features
- 📊 **Correlation Matrix:** Identify relationships between features.

PHASE 3: PREPROCESSING STRATEGY

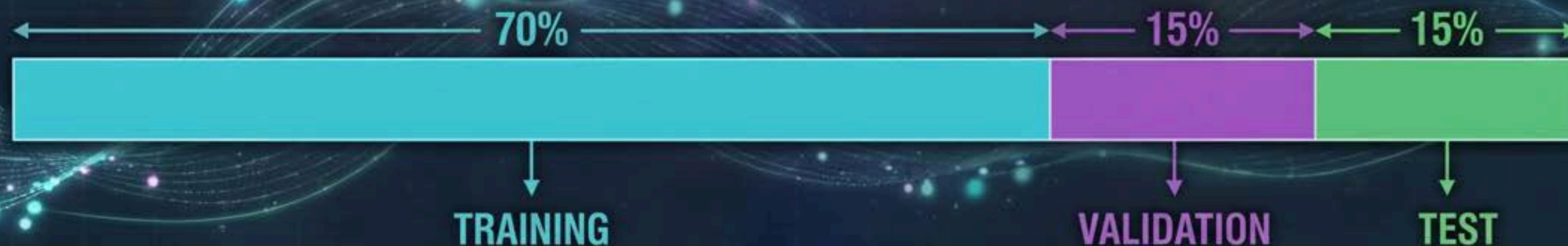
STANDARDIZATION (Z-SCORE)



Transform variables to Mean 0, Std Dev 1.

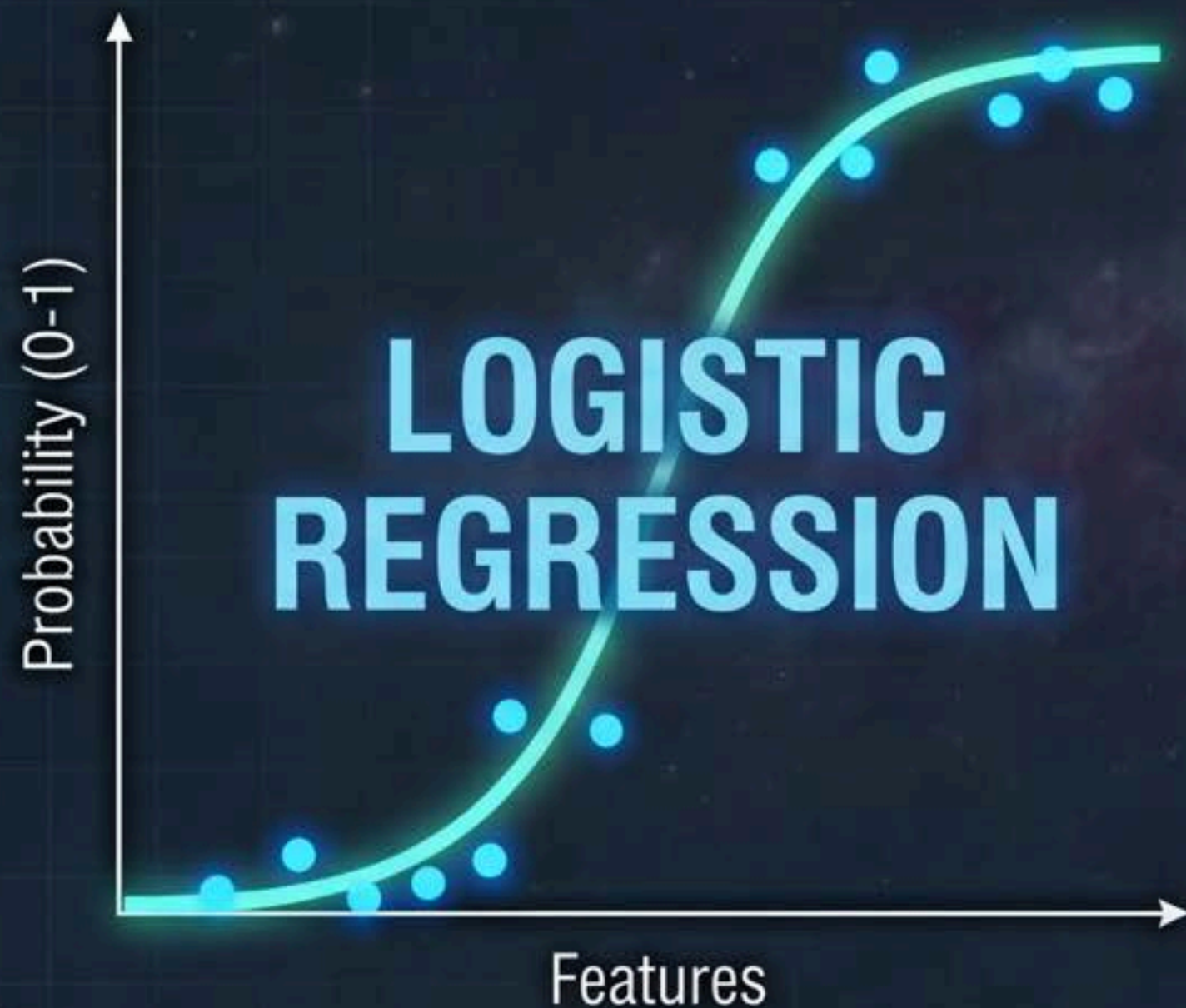
Rationale: Prevents scale disparity (Millions of km vs. km/s) from destabilizing the loss function.

DATASET SPLIT



PHASE 4: BASELINE MODEL

LOGISTIC REGRESSION



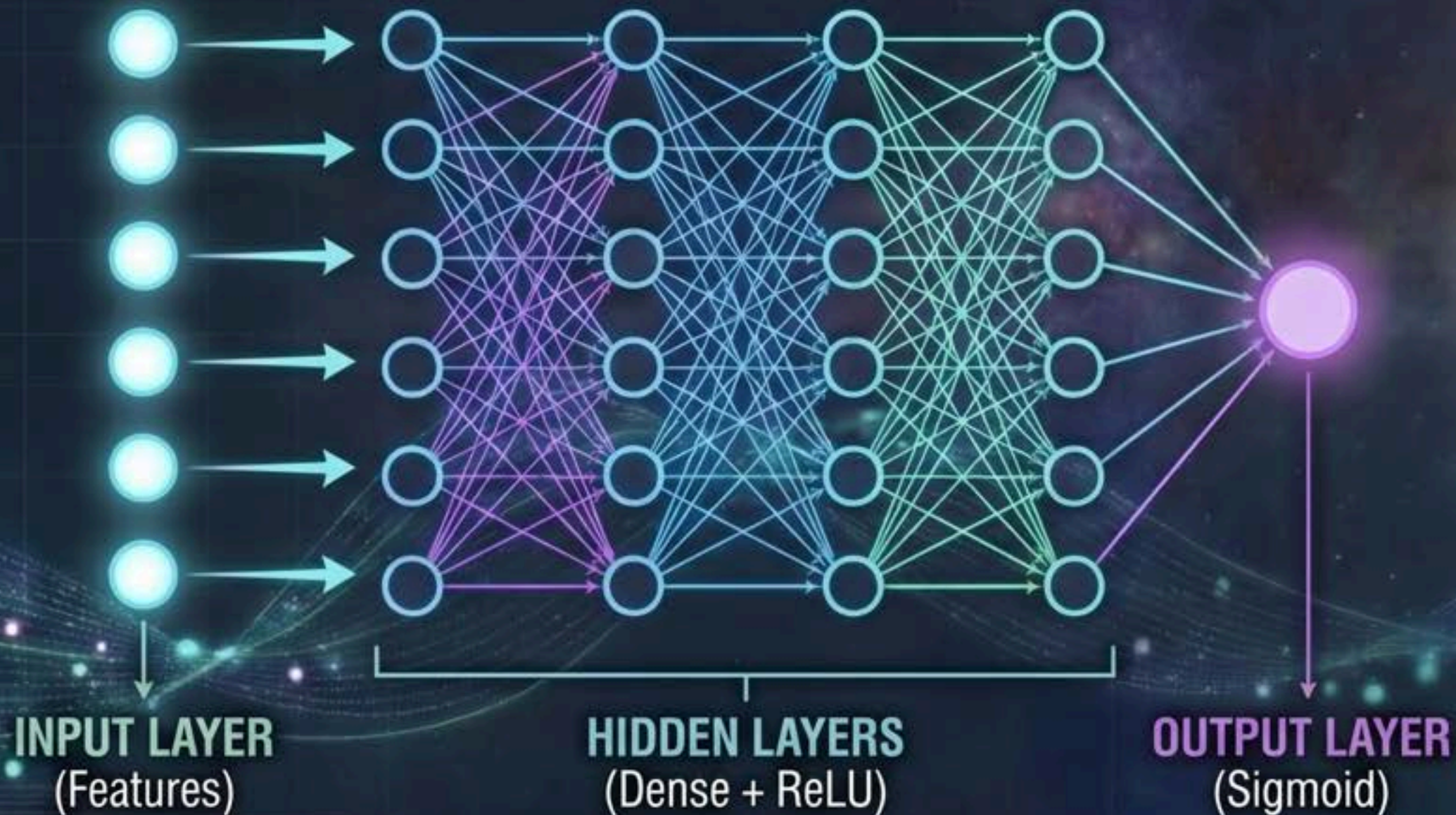
STRATEGIC PURPOSE: ESTABLISHING THE BENCHMARK

Algorithm: Logistic Regression

Logic: Simple, interpretable, low-cost

Criterion: The Neural Network must significantly outperform this baseline to justify its computational cost

PHASE 5: ADVANCED MODEL (ANN ARCHITECTURE)



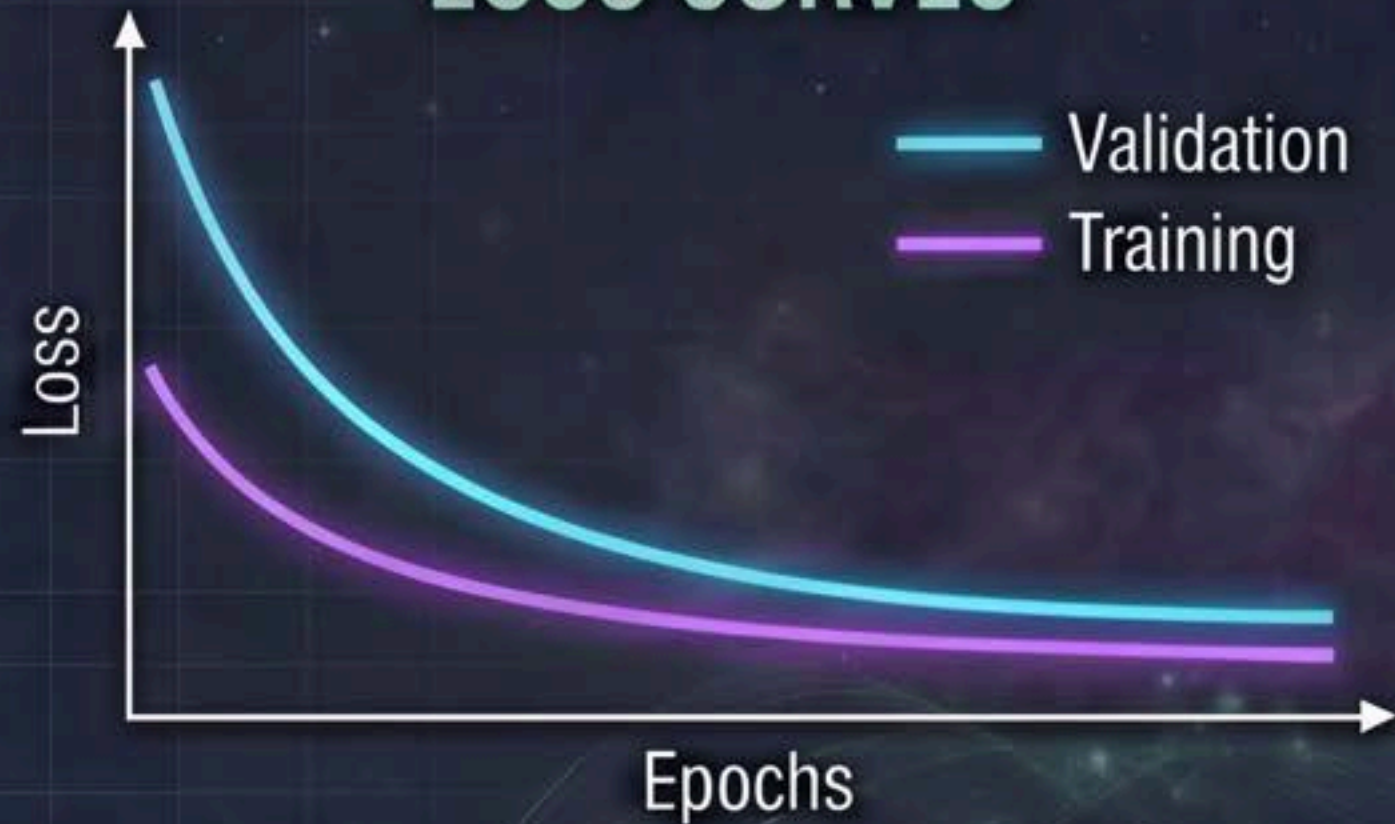
FRAMEWORK:
TENSORFLOW / KERAS

**COMPILATION
PARAMETERS:**

- **Optimizer:** Adam (Adaptive Moment Estimation)
- **Loss Function:** binary_crossentropy
- **Output:** Probability Score (0.0 - 1.0)

PHASE 6: EVALUATION

LOSS CURVES



Check for Overfitting/Underfitting

CONFUSION MATRIX

True Negative (Safe)	False Positive (False Alarm)
False Negative (MISSED THREAT) ⚠	True Positive (Detected)

CRITICAL METRIC: RECALL

Priority is minimizing False Negatives. Missing a hazardous asteroid is the worst-case scenario.

INTELLIGENCE RESOURCES

PRIMARY DATASET



🔍 NASA Nearest Earth Objects

Kaggle Repository: NASA Nearest Earth Objects

OFFICIAL MONITORING

CNEOS (Center for Near Earth Object Studies)

PDCO (Planetary Defense Coordination Office)

ACADEMIC REFERENCE

IEEE Xplore: "Classification and Comparative Analysis of Earth's Nearest Objects using Machine Learning Models."

CASE STUDIES

DART Mission (2022) success

Apophis (2029) potential threat

Chelyabinsk Event (2013)

Tunguska Event (1908)

MISSION TEAM

POLITECNICO DI BARI
Dipartimento di Meccanica, Matematica e Management
Course: Big Data Analytics

PROJECT OPERATORS:
Fabrizio M. Crimi
Michele Falconetti
Michele Storelli

STATUS: IN PROGRESS

